Statistical Mechanics (MP461) Assignment 2

Second assignment - hand in by 09.00am Friday - October 16th 2026. All starred ∗ questions to be attempted- and more if you like. If you have questions about this assignment, please ask me: (`graham-dot-kells-at-mu-dot-ie`), Office 1.12, Physics

*1. Molar masses

Calculate the mass of a mole of dry air, which is a mixture of $N_2(78\%)$, $O_2(21\%)$ and argon (1%).

*2. Heat capacity of water

The heat capacity of 0.125kg of water is measured to be 523J/K at room temperature.
(i) Use this to calculate the heat capacity of water per unit mass (e.g. the *specific* heat capacity).
(ii) What is the molar mass of water? Use this to calculate the *molar* heat capacity of water.

3. Heat Exchange

Two bodies, with heat capacities C_1 and C_2 (assumed independent of temperature) and initial temperatures T_1 and T_2 respectively, are placed in thermal contact. Show that their final temperature T_f is given by $T_f = (C_1T_1 + C_2T_2)/(C_1 + C_2)$. If C_1 is much larger than C_2, show that $T_f \approx T_1 + C_2(T_2 - T_1)/C_1$.

4. Exponential atmosphere

(a) Consider a horizontal slab of air whose thickness is dz. If this slab is at rest, the pressure holding it up below must balance both the pressure from above and the weight of the slab. Use this fact to find an expression for dp/dz, the variation of pressure with altitude in terms of the density of the air.

(b) Use the ideal gas law to write the density of air in terms of pressure, temperature, and the average mass m of the air - see Problem 1 above. Show then that the pressure obeys the barometric equation

$$\frac{dp}{dz} = -\frac{mg}{k_BT}p \tag{1}$$

(c) Assuming that the temperature of the atmosphere is independent of height (not a great assumption but not terrible either), solve the barometric equation to obtain the pressure as a function of height.

(d) Find a similar expression for the number density n as a function of height.

***5. Better calculations through Boltzmann**

Use your understanding of Boltzmann factors to come up with the same expression [as Q4 (d)] for $n(z)$.

***6. A three-level atom**

An atom can be in one of three states, with energies 0 eV, 0.2 eV and 0.4 eV. The atom is at thermal equilibrium with a reservoir at 2500 K. Calculate (a) the probability for this atom to be in each of the three states, and (b) the average energy of the atom.

7. Macrostates and microstates

A system consists of a set of uniformly spaced energy levels of separation $1\times10^{-20}J$, starting from the first level which has an energy of $0J$. Eight distinguishable particles are placed within this system with a total energy of $6 \times 10^{-20}J$.

(a) How many possible configurations are there satisfying this condition?
(b) Which of these is the most probable?

8. A question of speed

Show that for a Maxwell-Boltzmann velocity distribution $v_{max} < \langle v \rangle < v_{rms}$.

***9. Equipartition Theorem**

(a) If the energy of a classical system is a sum of n quadratic modes, i.e.

$$E = \sum_{i=1}^{n} \alpha_i x_i^2, \tag{2}$$

show that if the system is in contact with a heat reservoir at temperature T, the mean energy of the system is given by

$$\langle E \rangle = n \times \frac{1}{2} k_B T \tag{3}$$

10.Heat capacity and equipartition

Consider a crystalline solid consisting of N particles of mass m. Use the equipartition theorem to derive that, at sufficiently high temperatures, the molar specific heat of such a solid equals 3R, where R is the gas constant.